

HLA regions from 610 assembled haplotypes: extraction, typing and pangenome views

Robert Hoehndorf (KAUST)

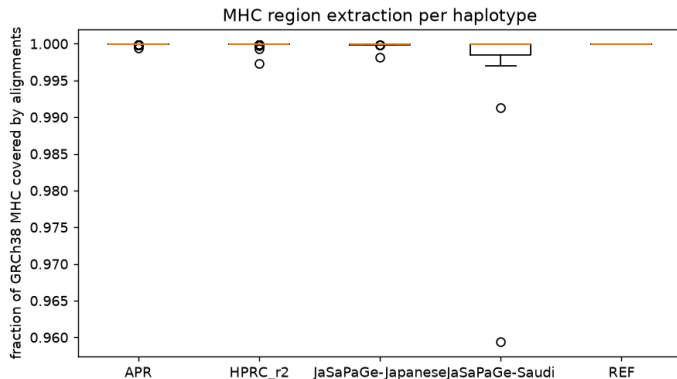
BioHackathon Japan 2026, HLA/pangenome group

15 September 2026

One CWL workflow: pgr-tk extraction, Immuannot typing, pgr-tk and pggg graphs

- ▶ 610 haplotypes: APR (UAE Arab) 106, HPRC release 2 464, JaSaPaGe Saudi 18 and Japanese 20, GRCh38, CHM13
- ▶ MHC (GRCh38 chr6:28.5–33.5 Mb, 100 kb flank) pulled from each assembly with pgr-tk pgr-query, about 30 s per genome
- ▶ Immuannot (IPD-IMGT/HLA 3.55) gives gene structure and full-resolution allele calls per haplotype
- ▶ Per gene (19 genes): sequences cut at the Immuannot coordinates, pgr-tk principal-bundle plots, pggg graphs rendered with odgi
- ▶ Whole MHC and the class II block (DRA to DMA): pgr-tk bundle decomposition, dendrograms, bundle graphs
- ▶ Runs on the NIG node without containers (Toil, --no-container); repo github.com/leechuck/pangenome-bh26

The MHC comes out intact from 608 of 610 haplotypes



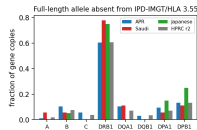
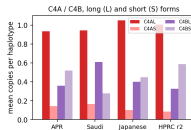
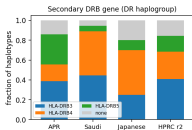
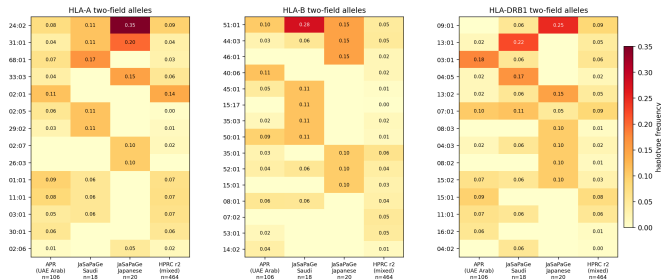
- y: fraction of the GRCh38 MHC covered by the extracted segments; median 1.000 in every cohort, 599 haplotypes on a single contig
- Five 1000G JPT individuals were assembled by both HPRC and JaSaPaGe: 59 of 61 full-resolution gene calls agree, the two exceptions differ only in an intron

How: MHC extraction with pgr-tk (workflow steps prepare_reference, extract_pgrtk)

- ▶ Query: GRCh38 chr6:28,510,120–33,480,577 plus 100 kb flank, cut from the PanSN GRCh38 fasta with samtools faidx
- ▶ Per assembly: `pgr-query --fastx-file -w 80 -k 56 -r 4 --min-span 64 --merge-range-tol 100000` (SHIMMER index of the whole assembly in memory, 8 threads, about 30 s and 9 GB per 3 Gb haplotype)
- ▶ Hits under 20 kb or under 20 anchors dropped; minus-strand hits reverse-complemented; records renamed `sample#hap#contig:start-end`; coverage = query span covered by kept hits (`scripts/pgr_query_rename.py`)
- ▶ Assemblies already PanSN-named (HPRC r2, references, JaSaPaGe NA*) are not prefixed twice; samples present in two projects get a `.cohort` suffix
- ▶ Concordance check: full-resolution Imm annot calls of the five JPT individuals compared between their HPRC and JaSaPaGe assemblies, per gene, as unordered genotypes

Allele frequencies separate the cohorts; DRB1 full-length sequences are mostly new

HLA allele landscape across 608 assembled haplotypes (Immuannot on assembly-derived MHC)

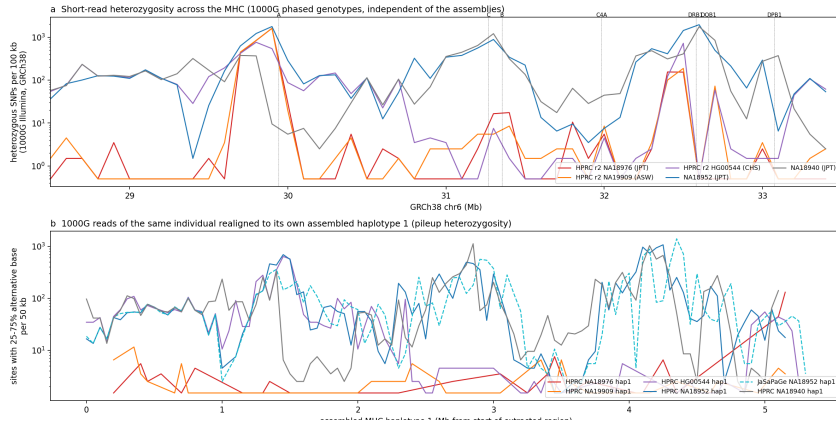


Immuannot consensus calls, two-field alleles; novel = full-length sequence absent from IPD-IMGT/HLA 3.55

How: allele calling and aggregation (steps immuannot, aggregate)

- ▶ Immuannot v3 (`immuannot.sh -c <mhc.fa> -r Data-2024Feb02 --diff 0.03, 4 threads`) on each extracted MHC: IPD-IMGT/HLA 3.55 gen sequences, IPD-KIR 2.13, RefSeq C4; 610 runs, about 2 h on 28 cores
- ▶ Allele = Immuannot consensus field of the transcript row; `:new` marks a full-length sequence absent from the database; novel coding vs noncoding from `cds_distance`
- ▶ `scripts/aggregate_immuannot.py`: `hla_calls.tsv` (one row per gene copy), haplotype x gene matrix, copy numbers (C4A/B long/short, DRB3/4/5 presence), per-cohort two-field frequencies and plots
- ▶ Cohorts: assembly folder plus file-name rule; JaSaPaGe split into Saudi (`ksa*`) and Japanese (`NA*`, 1000G JPT) in `scripts/make_inputs.py`
- ▶ Figure: `analysis/fig5_population.py`

Two HPRC individuals are truly MHC-homozygous; one JaSaPaGe assembly carries the same haplotype twice



- NA18976, NA19909 (red, orange): under 100 heterozygous 1000G SNPs across 5 Mb, matching their two near-identical assembled haplotypes; JaSaPaGe NA18952 (dashed): own reads show full heterozygosity on both “haplotypes”, hap1 = hap2

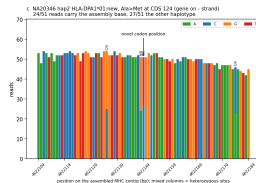
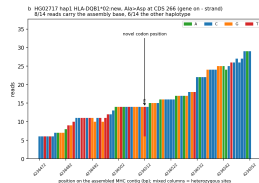
How: assembly-independent and read-level checks (scripts/read_support.sh, analysis/fig3)

- ▶ Candidates: individuals whose two haplotypes carry identical alleles at all eight classical genes (NA18976, NA19909, JaSaPaGe NA18952; HG00544 at seven)
- ▶ Panel a: 1000G high-coverage phased VCF (chr6, 3,202 samples), heterozygous SNPs per 100 kb for the candidates and controls (bcftools view -r chr6:28.51-33.48M)
- ▶ Panel b: MHC-region reads (chr6:28.4–33.6 Mb) pulled from the public 1000G CRAMs (samtools view on the EBI URL with the GRCh38 chr6 reference), realigned to the individual's own assembled haplotype 1 with minimap2 -ax sr; samtools mpileup, sites with 25–75% alternative base at depth ≥ 15 , counted per 50 kb
- ▶ Panel c: minimap2 -cx asm20 -cs of haplotype 2 against haplotype 1 of the same individual; substitutions per 50 kb from the cs tag
- ▶ JaSaPaGe NA18952 hap2 file carries the HPRC contig name JBHIJT010000011.1; hap1 (hifiasm contig) has the same alleles: the assembly holds one haplotype twice

Substitution-type novel coding alleles hold up in the individual's own reads; indel-only ones do not

a Candidate novel coding alleles (ImmunoNot 'new' with CDS differences) in classical HLA genes. private = variant 31-mer absent from all other 609 haplotypes; red = differences include indels (typical HiFi homopolymer assembly errors, treat as artefacts); green = heterozygous support in the individual's own short reads

haplotype	cohort	gene	closest known allele	coding difference(s) (known > assembly)	private variants	also on other hap	own 1000G Illumina reads (realigned to assembly)
apr003#1	APR	A	A*24:02:102:01	4 subst.: Val>Leu, Phe>Tyr, Ser>Pro, Lys>Asn	1	no	no public reads
apr011#1	APR	C	C*03:02:02:01	3 subst.: Ile>Leu, Ala>Gly, Gly>Cys	0	no	no public reads
apr001#1	APR	DRB1	DRB1*16:01:01:01	5 subst.: Gln>His, Val>Val, Leu>Arg, Met>Val, Arg>Arg	0	no	no public reads
apr038#1	APR	DRB1	DRB1*14:06:01	5 subst. + indels: Arg>Lys, Val>Ala, Arg>Arg, Cys>Cys, His>Gln	1	no	no public reads
apr002#1	APR	DQB1	DQB1*02:02:01:01	1 subst.: Leu>Leu	0	yes	no public reads
apr015#2	APR	DPB1	DPB1*10:01:01:01	1 subst.: Leu>Leu	1	no	no public reads
HG02717#1	HPRC r2	DQB1	DQB1*02:02:01:01	1 subst.: Ala>Asp	0	no	8/14 assembly base, 6 other hap
NA20346#2	HPRC r2	DPA1	DPA1*01:03:01:18Q	1 subst.: Ala>Met	1	no	24/51 assembly base, 27 other hap
ksa004#1	JaSaPaGe-Saudi	A	A*02:01:01:134Q	3 indels only (1-bp deletion, 2 insertions)	-	-	no public reads
HG03516#2	HPRC r2	DPA1	DPA1*03:01:01:01	1-bp insertion only	-	-	-

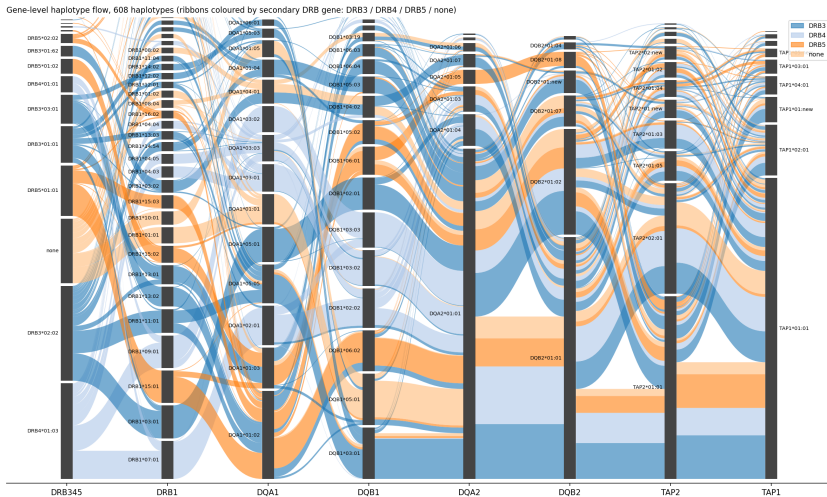


Pileups: the individual's 1000G Illumina reads realigned to its assembled contig; mixed columns = heterozygous sites. APR and JaSaPaGe reads are not public

How: novel-allele support (scripts/novel_alleles.py, analysis/fig4)

- ▶ Immuanot cds_mut gives, per transcript, the closest allele and the CDS-relative differences (cs-style: N matching bases then a substitution, assembly base first); converted to contig coordinates through the CDS rows, strand-aware
- ▶ Independent-assembly support: the 31-mer centred on each variant (and its reverse complement) searched in the same gene cut from all other 609 haplotypes; count of carriers, and whether the other haplotype of the same individual carries it
- ▶ Read support (HPRC individuals only): 1000G Illumina reads of the same person realigned to the assembled contig (minimap2 -ax sr), samtools mpileup ± 40 bp around the novel codon, per-base counts
- ▶ Indel-only “new” alleles (ksa004 HLA-A, HG03516 DPA1) flagged as likely HiFi homopolymer errors; APR raw reads are not public (SharePoint holds only DeepVariant VCFs)

425 distinct gene-level class II haplotypes among 608; secondary DRB gene structures the flow

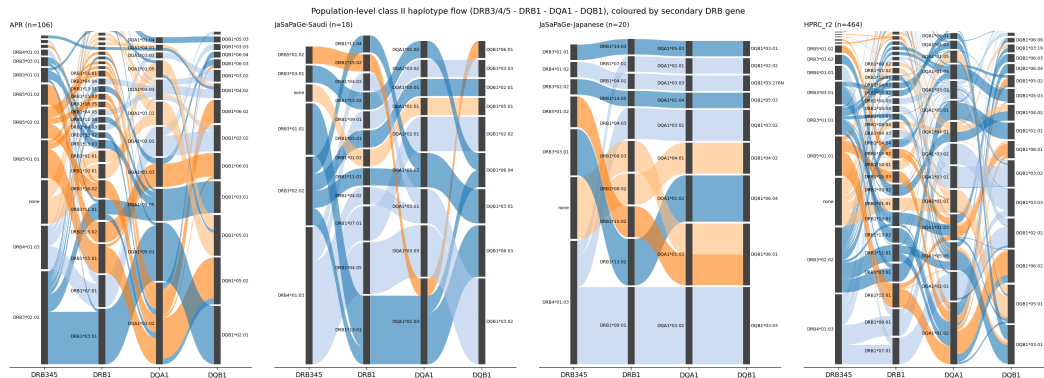


Two-field alleles per haplotype; ribbons coloured by DRB3 / DRB4 / DRB5 / none. After Chin, ASHI 2023 (94 haplotypes)

How: class II gene-level haplotype strings and flow plots (analysis/classII_flow_pca.py)

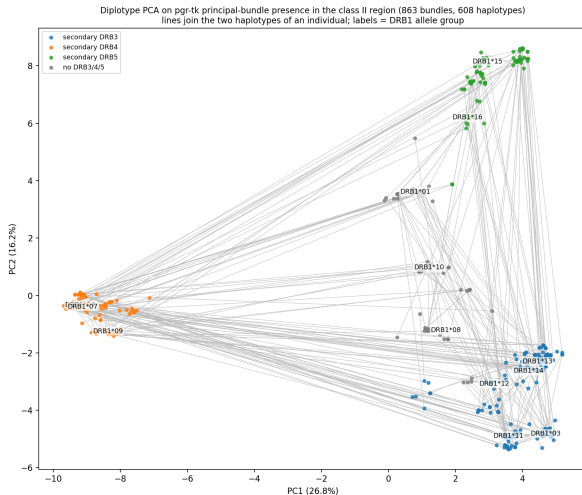
- ▶ Per haplotype: two-field Immuannot allele of DRA, DRB3/4/5 (collapsed into one column, “none” if absent), DRB1, DQA1, DQB1, DQA2, DQB2, DOB, TAP2, TAP1, DMB, DMA, joined into one string
(results/tables/classII_haplotype_strings.tsv)
- ▶ Alluvial plot: columns are genes, node height = haplotypes carrying the allele, ribbons = haplotypes shared between adjacent alleles, coloured by secondary DRB gene; drawn with matplotlib paths
- ▶ Per-cohort version uses the same code on the DRB3/4/5–DRB1–DQA1–DQB1 columns of each cohort
- ▶ Diplotype PCA (fig 8): one-hot encoding of the six DR/DQ alleles, centred, SVD; both haplotypes of an individual joined by a line

The DR-DQ block differs by population



- Saudi: DRB4 haplotypes (DR7, DR4) 44%; Japanese: DRB1*09:01–DQB1*03:03 and DRB1*15:02–DQB1*06:01; APR: DRB1*03:01–DQB1*02:01 18%

Sequence-level structure of the class II region follows the DR haplogroups

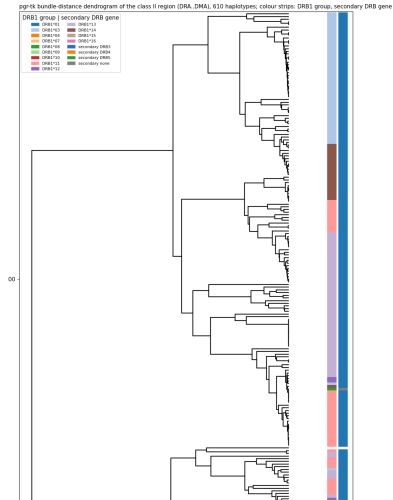


PCA on presence of 863 pgr-tk principal bundles (DRA to DMA); each point one haplotype, grey lines join the two haplotypes of an individual

How: class II bundle decomposition with pgr-tk (scripts/extract_span.py, tools/pgr_pbundle.cwl, analysis/classII_bundles.py); also slides 10 and 11

- ▶ Span: from HLA-DRA to HLA-DMA (outermost Immuannot coordinates, 20 kb flank) cut from every MHC haplotype in DRA orientation with samtools faidx; 610 sequences, 461–650 kb
- ▶ `pgr-pbundle-decomp -w 48 -k 56 -r 2 --min-span 8 --min-branch-size 8 --bundle-length-cutoff 500 --bundle-merge-distance 2000`: 863 principal bundles;
`pgr-pbundle-bed2dist` for the distance matrix and dendrogram, `bed2sorted` for the bundle-presence vectors, `bed2svg` for the bundle plot (`classII.svg/html`)
- ▶ PCA (this slide): 608 x 863 bundle presence matrix, centred, SVD; colour = secondary DRB gene from Immuannot
- ▶ Dendrogram (slide 10): Newick from `bed2dist`, leaves mapped to haplotypes, colour strips from the allele calls; cluster purity at 10–80 clusters in `classII_cluster_purity.tsv`
- ▶ Bundle graph (slide 11): nodes = bundles from the bed file, edges = consecutive bundles along each haplotype weighted by haplotype count, Kamada-Kawai layout

Bundle-distance clusters are near-pure DRB1 allele groups

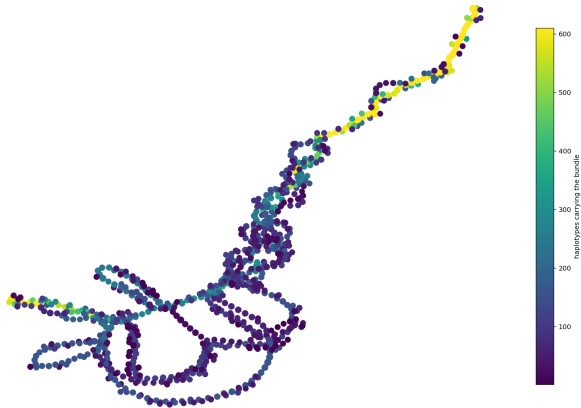


Top of the pgr-tk bundle-distance dendrogram of 610 class II haplotypes; strips: DRB1 group, secondary DRB gene. Full tree:

fig9_classII_dendrogram.png

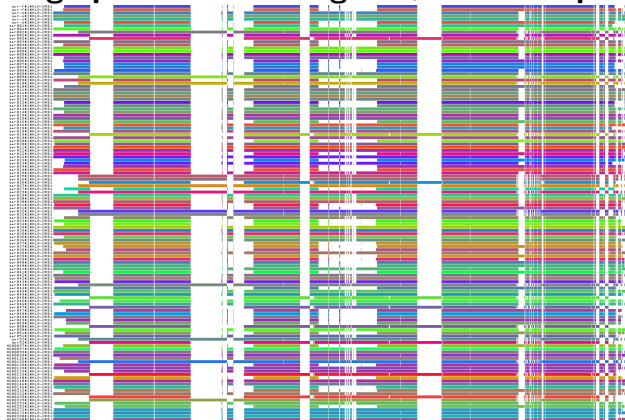
The class II bundle graph is linear except for the DRB block

pgprtk principal-bundle graph of the class II region (DRA, DMA), 630 haplotypes, 863 bundles, 1341 adjacencies
node size = bundle length, colour = number of haplotypes; edge width = haplotypes sharing the adjacency



863 principal bundles, node size = bundle length, colour = haplotypes carrying it, edges = consecutive bundles on a haplotype

Per-gene pangenome graphs for 19 HLA genes, all 610 haplotypes



- ▶ HLA-DRB1 pggb graph (odgi viz, top rows of 610): rows are haplotypes, x is the graph, colour = path coverage
- ▶ Also per gene: pgr-tk bundle plot (SVG/HTML), dendrogram, MAP-graph GFA; whole-MHC bundle plot

How: per-gene sequence sets and graphs (steps `gene_extract`, `gene_bundle`, `gene_graph`, `gene_viz`, `gene_draw`)

- ▶ Gene sequences: every annotated copy cut at its Immuannot coordinates with 2 kb flank, gene orientation, named `sample#hap#GENE` (`scripts/extract_genes.py`); a pgr-query homology fetch is available as alternative but needs denser seeds ($w=24/k=32$; DRB1 $w=16/k=24$) than the defaults
- ▶ pgr-tk per gene: `pgr-pbundle-decomp -w 48 -k 56 -r 2 --bundle-length-cutoff 200 --bundle-merge-distance 1000`, then `bed2dist / bed2sorted / bed2svg`
- ▶ pggb 0.7.4 per gene: `-p 90 -s 2000 -n <sequences> -P asm20`; odgi viz (1D, rows = haplotypes) and odgi layout + draw (2D)
- ▶ Whole MHC: `pgr-pbundle-decomp w=80, k=56, r=6, min_span 28` on the 3 Gb MHC fasta (about 40 min); optional whole-MHC pggb graph not run
- ▶ Runtime: Toil single-machine on 28 cores; extraction, annotation and graphs about 4 h in total

Take-home

- ▶ Assembly-based HLA typing across three pangenome projects reproduces known population frequencies and reveals data problems (duplicated haplotype, homozygous individuals) that short reads alone would not
- ▶ 60–78% of DRB1 gene copies have no full-length match in IPD-IMGT/HLA; Arab haplotypes (APR + Saudi, 124) are a submission-ready resource once read support is added
- ▶ pgr-tk bundle decomposition recovers DR haplogroup structure without any allele calls; scales from 94 to 610 haplotypes
- ▶ Open: APR and JaSaPaGe raw reads for novel-allele validation; KIR and IGH with the same workflow; HPRC per-population split
- ▶ Workflow, scripts, figures and tables: github.com/leechuck/pangenome-bh26 (h1a/)